

① C is convex $\Rightarrow \theta_1 x_1 + \theta_2 x_2 \in C$ (for any $\theta_1, \theta_2 \geq 0$, $\theta_1 + \theta_2 = 1$)

② Assume $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in C$ ($k \geq 2$)
~~and~~ s.t. $\theta_1, \dots, \theta_k \geq 0$ and $\sum_{i=1}^k \theta_i = 1$

③ need to proof:

$$\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k + \theta_{k+1} x_{k+1} \in C$$

s.t. $\theta_1, \dots, \theta_{k+1} \geq 0$ and $\sum_{i=1}^{k+1} \theta_i = 1$

- Denote $\alpha = \sum_{i=1}^k \theta_i$, $\alpha + \theta_{k+1} = 1 \Rightarrow \alpha = 1 - \theta_{k+1}$

- define $w_i = \frac{\theta_i}{\alpha}$ for $i = 1 \dots k$, so $\sum_{i=1}^k w_i = \frac{\sum_{i=1}^k \theta_i}{\alpha} = 1$

defines $y = w_1 x_1 + w_2 x_2 + \dots + w_k x_k$, s.t. $w_i \geq 0$ and $\sum_{i=1}^k w_i = 1$

According to ②, $y \in C$

Totally $\alpha y = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$

for $\alpha y + \theta_{k+1} x_{k+1}$, $\alpha + \theta_{k+1} = 1$, $y \in C$, $x_{k+1} \in C$

$\therefore \alpha y + \theta_{k+1} x_{k+1} \in C$



Q2: Necessity: C is convex $\rightarrow$ intersections with any line is convex

Proof: ① $C \subseteq \mathbb{R}^n$ is convex

$$\theta x_1 + (1-\theta)x_2 \in C \quad (s.t. \theta \in [0,1])$$

② $L \triangleq \{x \in \mathbb{R}^n \mid x = p + td, t \in \mathbb{R}\}$ (p is a point, $d \in \mathbb{R}^n, t \in \mathbb{R}$)

③ $C \cap L \triangleq \{x \in L \mid x \in C\}$

for any $x_1, x_2 \in C \cap L$, like $x_1 = p + t_1 d, x_2 = p + t_2 d$

$$\theta x_1 + (1-\theta)x_2 \in C$$

$$\theta(p + t_1 d) + (1-\theta)(p + t_2 d) = p + [\theta t_1 + (1-\theta)t_2]d \in L$$

$$\therefore \theta x_1 + (1-\theta)x_2 \in C \cap L$$

$\therefore$ Necessity is proof

Sufficiency: intersection of C with any line is convex

$\rightarrow C$ is convex

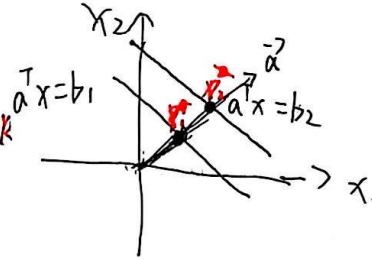
Proof: ① Assume $C \cap L$ is convex

$$\theta x_1 + (1-\theta)x_2 \in C \cap L \in C$$

$\therefore$ sufficiency is proof

Q3: distance between two parallel hyperplanes

$$a^T x = b_1, \quad a^T x = b_2$$



$$① \quad p_1 = \theta \vec{a} \quad p_2 = \beta \vec{a}$$

$$② \quad \begin{cases} a \cdot p_1 = b_1 \\ a \cdot p_2 = b_2 \end{cases}$$

① substitute into ②,

$$\begin{cases} \theta = b_1 / a^T a \\ \beta = b_2 / a^T a \end{cases}$$

$$\therefore \text{distance} = |p_2 - p_1| = (\beta - \theta) \vec{a} = \frac{|b_2 - b_1|}{\|a\|}$$

Q4

a) $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$ where $a_1, a_2 \in \mathbb{R}^n$

Convex

Proof: ① $z_1 = y_1 a_1 + y_2 a_2, z_2 = y'_1 a_1 + y'_2 a_2$ ($s.t. -1 \leq y_1, y'_1 \leq 1, -1 \leq y_2, y'_2 \leq 1$)

$$② \quad \theta z_1 + (1-\theta)z_2 = \theta(y_1 a_1 + y_2 a_2) + (1-\theta)(y'_1 a_1 + y'_2 a_2), \theta \in [0,1]$$

$$= [\theta y_1 + (1-\theta)y'_1]a_1 + [\theta y_2 + (1-\theta)y'_2]a_2$$

$$③ \quad \text{Since } -1 \leq y_1, y'_1 \leq 1 \Rightarrow [\theta y_1 + (1-\theta)y'_1] \in [-1, 1],$$

$$\Rightarrow [\theta y_2 + (1-\theta)y'_2] \in [-1, 1]$$

$$\therefore \theta z_1 + (1-\theta)z_2 \in S \Rightarrow S \text{ is convex}$$

b) $S = \{x \in \mathbb{R}^n \mid x \geq 0, 1^T x = 1, \sum_{i=1}^n a_i x_i = b_1, \sum_{i=1}^n b_i x_i = b_2\}$

convex ① $x \geq 0$ defines a convex region.

② $1^T x = 1, \sum_{i=1}^n a_i x_i = b_1, \sum_{i=1}^n b_i x_i = b_2$ are linear equality constraints, the solution space of a linear equality is convex

③ The intersection of ①, ② is convex, $\Rightarrow S$ is convex



(c) $S = \{x \in \mathbb{R}^n \mid x \geq 0, x^T y \leq 1 \text{ for all } y \text{ with } \|y\|_2 = 1\}$

(convex)

① Let $x_1, x_2 \in S$, for all y with $\|y\|_2 = 1$
 $x_1^T y \leq 1$ and $x_2^T y \leq 1$

② for $x = \theta x_1 + (1-\theta)x_2$

$$x^T y = [\theta x_1 + (1-\theta)x_2]^T y = \theta(x_1^T y) + (1-\theta)(x_2^T y) \\ \leq \theta + (1-\theta) \cdot 1 = 1$$

$\therefore x \in S$, S is convex

(d) $S = \{x \in \mathbb{R}^n \mid x \geq 0, x^T y \leq 1 \text{ for all } y \text{ with } \sum_{i=1}^n |y_i| = 1\}$

① Let $x = \theta x_1 + (1-\theta)x_2$, $\therefore x_1^T y \leq 1$ and $x_2^T y \leq 1$

$$x^T y = \theta(x_1^T y) + (1-\theta)(x_2^T y)$$

$$\leq \theta + (1-\theta) \cdot 1$$

$\therefore S$ is convex too

Q 6

(a) slab: $S = \{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}$

(convex) ① $x = \theta x_1 + (1-\theta)x_2$ s.t. $\theta \in [0, 1]$

$$a^T x = a^T [\theta x_1 + (1-\theta)x_2] = \theta a^T x_1 + (1-\theta)a^T x_2$$

Since $\alpha \leq a^T x_1 \leq \beta$ and $\alpha \leq a^T x_2 \leq \beta$

$$\Rightarrow \theta \alpha \leq \theta a^T x_1 \leq \theta \beta, (1-\theta)\alpha \leq (1-\theta)a^T x_2 \leq (1-\theta)\beta$$

$$\therefore \alpha \leq \theta a^T x_1 + (1-\theta)a^T x_2 \leq \beta$$

$\therefore x \in S$, S is convex

(b) $S = \{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i=1, \dots, n\}$

① Let $x_1, x_2 \in S$,

$$\Rightarrow \alpha_i \leq (x_1)_i \leq \beta_i \text{ and } \alpha_i \leq (x_2)_i \leq \beta_i$$

② $x = \theta x_1 + (1-\theta)x_2$ where $\theta \in [0, 1]$

$$x_i = \theta(x_1)_i + (1-\theta)(x_2)_i \quad \forall i=1, \dots, n$$

$$\alpha_i \leq x_i = \theta(x_1)_i + (1-\theta)(x_2)_i \leq \beta_i$$

$\Rightarrow S$ is convex

(c) $S = \{x \in \mathbb{R}^n \mid a_1^T x \leq b_1, a_2^T x \leq b_2\}$

① Let $x_1, x_2 \in S$, $a_1^T x_1 \leq b_1, a_2^T x_1 \leq b_2$
 $a_1^T x_2 \leq b_1, a_2^T x_2 \leq b_2$

② $x = \theta x_1 + (1-\theta)x_2$

$$a_1^T x = \theta a_1^T x_1 + (1-\theta)a_1^T x_2 \leq \theta b_1 + (1-\theta)b_1 = b_1$$

$$a_2^T x = \theta a_2^T x_1 + (1-\theta)a_2^T x_2 \leq \theta b_2 + (1-\theta)b_2 = b_2$$

$\therefore S$ is convex

